

ADVANCED CROP AND PLANT MONITORING SYSTEM WITH THE ASSISTANCE OF ARTIFICIAL INTELLIGENCE

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

This project introduces an AI-Powered Plant Monitoring System that combines real-time environmental tracking, AI-driven weed detection, and an interactive facial display to enhance plant care. The system uses sensors to monitor key parameters such as soil moisture, temperature, humidity, and light intensity, ensuring optimal growth conditions. The collected data is transmitted to a mobile application, providing users with daily insights, alerts, and care recommendations. A key feature is the AI-powered weed detection system, utilizing YOLO or TensorFlow-based image processing. A camera module captures images periodically, distinguishing between plants and invasive weeds. If weeds are detected, the system alerts the user through the app, enabling timely removal. Additionally, the system includes a facial display embedded in the vase or flower pot, acting as an expressive interface to reflect the plant's health. The display changes expressions—a happy face for optimal conditions, a sad face for low water levels, and an alarmed face for critical issues like extreme temperatures or pest presence. This smart system integrates real-time monitoring, AI-based automation, and an engaging user interface, making plant care efficient, interactive, and accessible for all users, from gardening enthusiasts to beginners.

TABLE OF CONTENTS

LIST OF FIGURES

CHAPTERS	TITLE	PAGE NO
1.	INTRODUCTION	
	1.1 Introduction	1
	1.2Block Diagram	3
	1.3 Methodology	3
2.	LITERATURE REVIEW	
	2.1 Existing System	5
	2.2 Proposed System	7
3.	METHODOLOGY	
	3.1 ESP 32 / CAM 32	8
	3.1.1 ESP Module Components	9
	3.1.2 ESP Module Ports	10
	3.1.3 Specifications	12
	3.1.4 Reasons For Choosing ESP	14
	3.1.5 Weed Detection using ESP CAM 32	15
	3.2 3D Printed Base	16
	3.2.1 Reasons For design	17
	3.3 Soil Moisture Sensor	17
	3.4 Temperature Sensor (DHT11)	18
	3.5 LDR	20
	3.6 I2C LCD	21
	3.6.1 Reason for Display	22

3.6.2 Usage of Display	23
3.7 Lithium Ion Battery	23
3.8 Micro python for ESP - 32	25
3.8.1 About Micro python	25
3.8.2 Installing Micro python in windows10	25
3.8.3 How to program ESP - 32 using Micro python	27
3.9 Applications for the Internet of Things(IOT) data management	28
3.10 TensorFlow	28
3.11 Google Collab	30
3.12Applications for the Artificial intelligence for Weed Detection and Plant Disease	31
3.13 About Weed Detection (AI)	32
3.14 About Plant Disease (AI)	32
3.15 Internet Of Things (IOT) for Plant Monitoring	33
4. RESULTS AND DISCUSSIONS	34
CONCLUSIONS	37
REFERENCES	38

LIST OF FIGURES

FIGURE.NO	TITLE	PAGE.NO
1.1	Block Diagram	3
2.1	ESP – 32 Microcontroller	9
2.2	ESP CAM 32 Microcontroller	9
2.3	ESP – 32 Processor	9
2.4	ESP – 32 PIN Diagram	12
2.5	ESP CAM 32 PIN Diagram	12
2.6	ESP CAM 32 Live Image	15
2.7	3D Model (SolidWorks)	16
2.8	Soil-Moisture Sensor	17
2.9	Temperature Sensor (DHT11)	18
2.10	LDR Sensor	20
2.11	I2C Display	21
2.12	Li – ion Battery	23
2.13	Micro Python Logo	25
2.14	Thonny IDE	26
2.15	TensorFlow Logo	28
2.16	Google Colab Logo	30
2.17	Weed Detection (AI)	32
2.18	Plant Disease (AI)	32

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The **AI-Powered Plant Monitoring System** aims to revolutionize plant care by integrating real-time environmental monitoring, AI-based weed detection, and an interactive facial display. The primary objective is to develop a smart system that continuously tracks key plant health parameters such as soil moisture, temperature, humidity, and light intensity using various sensors. This data is collected and transmitted to a dedicated mobile application, allowing users to access daily insights, alerts, and personalized care recommendations to ensure optimal plant growth.

A crucial aspect of the system is the implementation of AI-based weed detection using YOLO or TensorFlow-based image processing. A camera module periodically captures images of the plant and its surroundings, analyzing them to detect the presence of weeds or unwanted plants. Upon detection, the system alerts the user through the app, enabling timely removal and preventing weeds from competing with the plant for nutrients and resources. This feature enhances the overall efficiency of plant maintenance by automating weed detection.

Additionally, the project introduces a facial display embedded in the vase or flower pot, providing an interactive and engaging way to monitor plant health. This display uses expressive facial animations to indicate the plant's condition—a happy face for optimal conditions, a sad face for water deficiency, and an alarmed face for critical issues such as extreme temperatures or pest presence. This feature makes plant care more intuitive, especially for users with little gardening experience.

Furthermore, the system leverages IoT-based communication, ensuring seamless data transfer between the sensors, processing unit, and mobile application. By combining AI, IoT, and an engaging user interface, this project aims to make plant care efficient, interactive, and accessible, benefiting home gardeners, urban farming enthusiasts, and commercial plant maintenance applications.

Enhanced Plant Health Monitoring: This system continuously tracks vital parameters such as soil moisture, temperature, humidity, and light intensity, ensuring plants receive optimal care and reducing the risk of under or overwatering.

AI-Based Weed Detection: The integration of YOLO or TensorFlow-based AI models helps identify and notify users about unwanted weed growth, preventing competition for nutrients and promoting healthier plant development.

User-Friendly Mobile Application: The real-time data transmission to a mobile app provides users with daily insights, alerts, and care recommendations, making plant maintenance easier and more efficient.

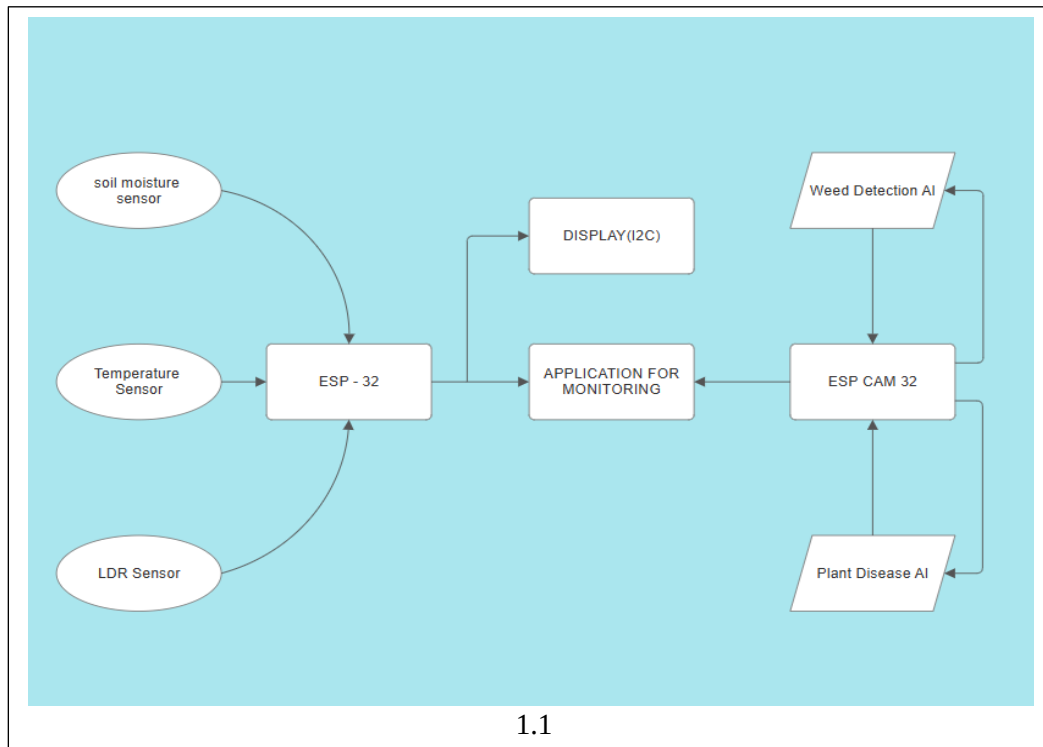
Interactive Facial Display: The embedded facial display in the vase or flower pot serves as an intuitive interface that visually represents the plant's health, making plant care more engaging and accessible, especially for beginners.

IoT-Enabled Smart System: The use of IoT technology ensures seamless data transfer between sensors, the processing unit, and the mobile app, enabling remote monitoring and control.

Automation and Reduced Manual Effort: By automating monitoring and weed detection, the system significantly reduces the need for constant manual supervision, making it ideal for busy individuals and commercial plant care applications.

Sustainability and Smart Gardening: The system promotes efficient water usage, early weed detection, and healthier plant growth, supporting sustainable urban gardening, smart agriculture, and eco-friendly plant management practices.

1.2 BLOCK DIAGRAM



1.3 METHODOLOGY

The AI-Powered Plant Monitoring System begins with real-time environmental data collection using a network of sensors that measure soil moisture, temperature, humidity, and light intensity. These sensors are connected to a microcontroller, which processes the collected data and transmits it to a cloud server or local database. A mobile application retrieves this data, providing users with daily insights, alerts, and recommendations for better plant care. The system ensures seamless IoT-based communication, allowing users to monitor plant health remotely and take timely action when necessary.

For weed detection, the system employs a camera module that captures periodic images of the plant's surroundings. These images are analyzed using AI-based deep learning models like YOLO or TensorFlow, which detect and classify weeds accurately. If weeds are detected, the system sends notifications to the user via the mobile app, enabling quick intervention to prevent competition for nutrients. Additionally, an interactive facial display embedded in the vase or flower pot visually represents the plant's condition, with expressions changing based on factors like moisture level, temperature stress, or pest presence. By integrating AI, IoT, and an engaging interface, this methodology ensures a smart, efficient, and user-friendly plant care system for gardening enthusiasts and commercial applications.

CHAPTER 2

LITERATURE REVIEW

2.1 Existing System:

Paper Title : An Effective Approach for Plant Monitoring, Classification and Prediction Using IoT and Machine Learning

Agriculture has been the most important sector of our country. As the world is growing and moving in a fast pace, and depending on the automation of most of the things there is also a need to maintain plants by doing some automation. In general, every plant has some specific needs that have to be addressed for its survival. Therefore, a system must be developed where plants can communicate with the user through IoT. By monitoring these parameters, we can ensure that the plants are healthy. There is also the need to analyze, collect and make the best use of the parameters for classifications about their state. In this project, the focus is mainly on two parts. The first part aims to model a system where we keep track of the requirements of the plants using sensors and IoT. Data collection is done through sensors and sent to the Blynk app. Automatic water controller activates only when the moisture content falls below a certain threshold. The second part aims to analyze the Blynk collected data to classify the plant based on its conditions, i.e. healthy or unhealthy. Finally, we compare the performance results of support vector machine, random forest and logistic regression for classification.

Paper Title : Design of An IoT based Smart Plant Monitoring System

In IoT plant monitoring system we can monitor and control with the help of IoT concepts. The gasses produced by the plant in Day time and nighttime will be monitored on IoT cloud. And according to the moisture of the soil the water pump supply to the plant will be controlled. If moisture is seen below the certain threshold value, then the pump will get turned on and when moisture reaches above the threshold level then the water pump will be turned off by the controller in the system. The Temperature and humidity will be sensed by smart sensors digitally. The hazardous gas will be sensed by the other sensor at that place. The controller which is integrated in the system will send all the monitored parameter and different gasses data on IoT Cloud with the help of Wi-Fi and IoT system. Other circuitry is used to control the relay and to provide power supply to the system. On the IoT cloud the data will be received on the google firebase Real-time database. From that it will be transferred to the AWS server and all the data will be visualized by different gauges and past data will be plotted.

2.2 Proposed System

The proposed plant monitoring system integrates temperature, moisture, and humidity sensors with an AI-powered model for weed detection and plant disease identification. Utilizing ESP32 and ESP32-CAM modules, the system collects real-time environmental and visual data to ensure effective crop monitoring. The ESP32 microcontroller gathers temperature, humidity, and soil moisture data, which is transmitted to a cloud platform such as Firebase or ThingSpeak for remote access. Simultaneously, the ESP32-CAM captures images of plants and sends them to an AI model based on YOLO for weed and disease detection. The AI model processes these images, identifies potential threats, and provides insights through a web or mobile application. Additionally, automated alerts notify users of unfavorable conditions, enabling timely action. The system can also include a relay module to automate irrigation when soil moisture levels are low, optimizing water usage. This smart plant monitoring system enhances agricultural productivity by providing real-time data, early disease detection, and efficient resource management, ensuring healthier crops and higher yields.

CHAPTER 3

METHODOLOGY

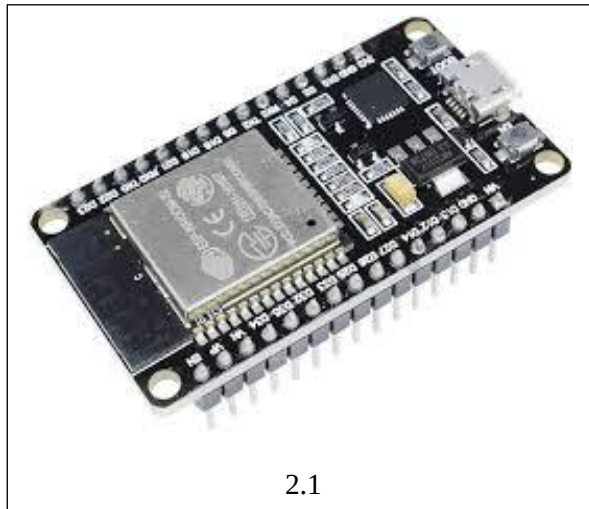
HARDWARE COMPONENTS:

3.1 ESP 32:

ESP32 is a series of low-cost, low-power system-on-chip microcontrollers with integrated Wi-Fi and dual-mode Bluetooth. The ESP32 series employs either a Tensilica Xtensa LX6 microprocessor in both dual-core and single-core variations, an Xtensa LX7 dual-core microprocessor, or a single-core RISC-V microprocessor and includes built-in antenna switches, RF balun, power amplifier, low-noise receive amplifier, filters, and power-management modules. It is commonly found either on device-specific PCBs or on a range of development boards with GPIO pins and various connectors depending on the model and manufacturer of the board.

ESP32 is created and developed by Espressif Systems, a Chinese company based in Shanghai, and is manufactured by TSMC using their 40 nm process. It is a successor to the ESP8266 microcontroller.

The ESP32-CAM is a compact, low-cost microcontroller with a built-in camera module, developed by Espressif Systems. It is based on the ESP32 chip and is widely used in IoT, AI-based image processing, and surveillance applications. With its Wi-Fi and Bluetooth capabilities, the ESP32-CAM can wirelessly transmit images and video, making it ideal for AI-powered projects like face recognition, object detection, and smart agriculture. It supports microSD card storage, allowing local image saving, and integrates with machine learning frameworks like TensorFlow Lite or OpenCV for real-time image analysis. Despite its small size and low power consumption, the ESP32-CAM provides high-performance processing, making it a popular choice for automated monitoring systems, security applications, and AI-driven plant health analysis.



2.1

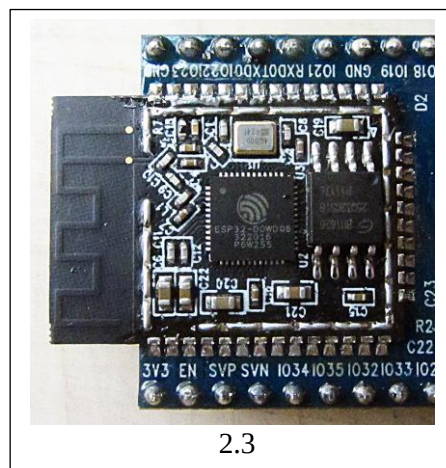


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3.1.1 ESP MODULE COMPONENTS:

The ESP32 module consists of several key components that enable its powerful performance for IoT, wireless communication, and automation.

1. **Dual-Core Processor (Xtensa LX6 CPU)** – The ESP32 features a dual-core 32-bit processor, providing high processing power for multitasking and real-time applications.



2.3

2. **Wi-Fi and Bluetooth Connectivity** – Built-in Wi-Fi (802.11 b/g/n) and Bluetooth (v4.2 and BLE) allow seamless wireless communication for IoT projects.
3. **GPIO (General-Purpose Input/Output) Pins** – The module includes multiple GPIO pins for interfacing with sensors, actuators, and peripherals like cameras, LEDs, and motors.
4. **Flash Memory and RAM** – The ESP32 comes with 4MB to 16MB of Flash memory for program storage and 520KB SRAM, enabling fast data processing.
5. **Power Management (Low Power Consumption)** – Features deep sleep and ultra-low power modes, making it energy-efficient for battery-operated applications.
6. **ADC & DAC (Analog and Digital Conversion)** – The ADC (Analog-to-Digital Converter) and DAC (Digital-to-Analog Converter) allow the ESP32 to handle analog signals for sensor data collection.
7. **Communication Interfaces** – The module supports multiple communication protocols, including SPI, I2C, UART, PWM, and I2S, enabling easy integration with various devices.
8. **Cryptographic Hardware Acceleration** – It has built-in encryption and security features, ensuring safe communication for IoT applications.

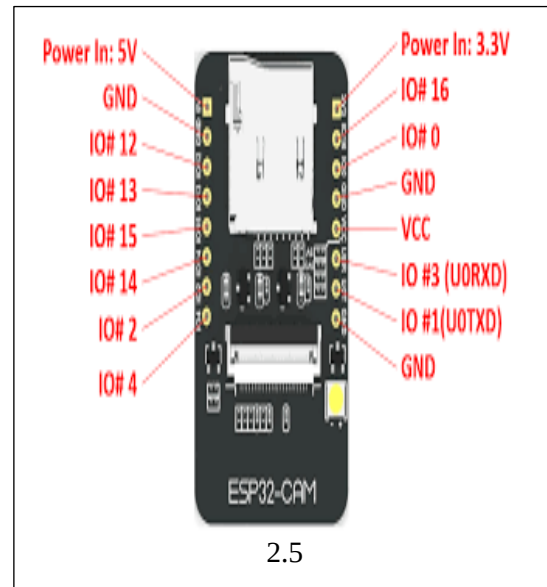
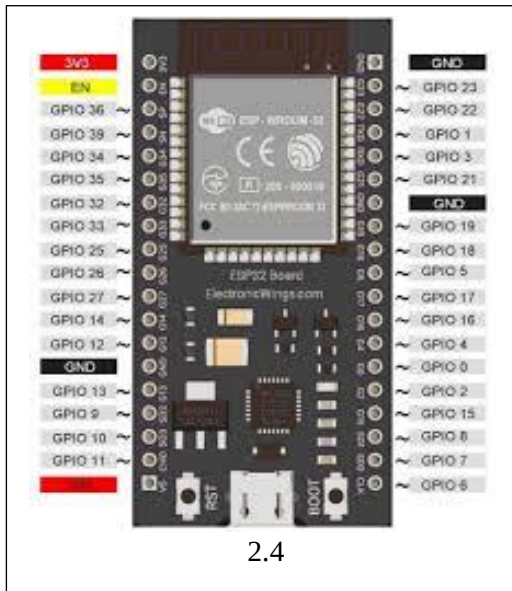
3.1.2 ESP MODULE PORTS:

The ESP32 module features multiple ports and interfaces that allow communication with sensors, actuators, and external devices. Below are the key ports available on the ESP32:

1. **Power Ports (VCC & GND)** – The VCC (3.3V or 5V) port supplies power to the ESP32, while GND (Ground) is used for circuit grounding.

2. **GPIO (General-Purpose Input/Output) Ports** – The ESP32 has up to 34 GPIO pins, which can be used for digital input and output to connect sensors, LEDs, motors, and other peripherals.
3. **Analog Input Ports (ADC Pins)** – The module includes 18 ADC (Analog-to-Digital Converter) channels, allowing it to read analog signals from temperature sensors, potentiometers, and other analog devices.
4. **DAC (Digital-to-Analog Converter) Ports** – The ESP32 has two DAC pins (GPIO25 & GPIO26) that allow it to generate analog signals, useful for applications like audio output and signal modulation.
5. **PWM (Pulse Width Modulation) Ports** – Almost all GPIO pins on the ESP32 support PWM, making it useful for controlling servo motors, LED brightness, and speed control in DC motors.
6. **Communication Ports:**
 - UART (Universal Asynchronous Receiver-Transmitter) Ports – The ESP32 has three UART interfaces (UART0, UART1, UART2) for serial communication with other microcontrollers, GPS modules, or external sensors.
 - I2C (Inter-Integrated Circuit) Ports – The ESP32 supports two I2C interfaces for communicating with LCD displays, sensors, and other peripherals. The default SDA (GPIO21) and SCL (GPIO22) pins can be reassigned as needed.
 - SPI (Serial Peripheral Interface) Ports – The ESP32 supports two SPI buses, commonly used for interfacing with SD cards, displays, and external memory modules.
7. **Touch Sensor Ports** – The ESP32 includes 10 capacitive touch sensor pins, enabling touch-based input without the need for mechanical buttons.
8. **JTAG Debugging Ports** – The ESP32 supports JTAG debugging, which allows developers to debug firmware and analyze code execution.

9. Micro USB Port (For Programming and Power) – Some ESP32 development boards include a Micro USB port for programming using the Arduino IDE, ESP-IDF, or Micro Python, as well as for providing power.



3.1.3 SPECIFICATIONS:

The ESP32 is a powerful and versatile microcontroller with advanced features for IoT, AI, and embedded system applications. Below are its key specifications:

1. Processor & Performance

- CPU: Dual-core Xtensa LX6 processor
- Clock Speed: Up to 240 MHz
- SRAM: 520 KB (Internal)
- ROM: 448 KB
- Flash Memory: 4 MB to 16 MB (varies by module)
- Ultra-Low Power Co-Processor: Handles low-power tasks without waking the main CPU

2. Wireless Connectivity

- Wi-Fi: 802.11 b/g/n (2.4 GHz) with speeds up to 150 Mbps
- Bluetooth: Bluetooth 4.2 and Bluetooth Low Energy (BLE)

3. Input/Output (I/O) Interfaces

- GPIO Pins: Up to 34 GPIOs
- Analog Input (ADC): 18 channels, 12-bit resolution
- Digital-to-Analog Converter (DAC): 2 channels, 8-bit resolution
- Touch Sensor Pins: 10 capacitive touch sensors
- PWM (Pulse Width Modulation): Available on most GPIOs

4. Communication Protocols

- UART: 3 UART interfaces (for serial communication)
- SPI: 4 SPI interfaces (for fast data exchange)
- I2C: 2 I2C interfaces (for connecting sensors and peripherals)
- I2S: For audio processing and sound applications
- CAN Bus: Supports Controller Area Network communication

5. Power & Consumption

- Operating Voltage: 3.3V
- Power Consumption Modes:
- Active Mode: ~160 mA
- Deep Sleep Mode: <10 μ A (for battery efficiency)
- Hibernate Mode: ~2.5 μ A
- Supports Battery & Solar Power

6. Security & Encryption

- Built-in Cryptographic Acceleration: AES, RSA, SHA, ECC
- Secure Boot & Flash Encryption

7. Operating Environment

- Operating Temperature: -40°C to 85°C
- Humidity Resistance: Suitable for outdoor IoT applications

8. Special Features

- Supports Over-the-Air (OTA) Updates
- JTAG Debugging Support
- AI & Machine Learning Compatibility (TensorFlow Lite, Edge Impulse)

3.1.4 REASONS FOR CHOOSING ESP MODULE:

The ESP32 module is chosen for this project due to its powerful performance, wireless connectivity, and versatility in IoT applications. It features a dual-core processor with a clock speed of up to 240 MHz, enabling efficient real-time processing of sensor data and AI-based image analysis. With built-in Wi-Fi and Bluetooth, the ESP32 allows seamless communication between the plant monitoring system and the mobile application, enabling remote data access and control. Its low power consumption and deep sleep mode make it an energy-efficient choice, especially for battery-operated or solar-powered applications. The ESP32 supports multiple communication protocols, including I2C, SPI, UART, and PWM, ensuring compatibility with various sensors, actuators, and peripherals. Additionally, the module's AI and machine learning capabilities allow integration with frameworks like TensorFlow Lite, making it suitable for weed detection and plant disease identification. The presence of ADC, DAC, and capacitive touch sensor pins further enhances its usability for various sensing and interactive applications. Its compact size, cost-effectiveness, and support for over-the-air (OTA) updates make the ESP32 an ideal microcontroller for developing a smart plant monitoring system with AI-driven automation and real-time data analytics.

3.1.5 WEED DETECTION USING ESP CAM - 32:

The ESP32-CAM is an ideal choice for AI-powered weed detection due to its built-in camera module, Wi-Fi connectivity, and compatibility with machine learning models. It captures real-time images of the plant environment, which are then processed using AI-based image recognition techniques like YOLO (You Only Look Once) or TensorFlow Lite to differentiate between weeds and healthy crops. The ESP32-CAM can be programmed to send alerts to a mobile app or cloud server when weeds are detected, allowing farmers or plant owners to take immediate action.

The module's compact size and low power consumption make it suitable for remote agricultural applications, where continuous monitoring is required. With Wi-Fi connectivity, it can transmit images and weed detection results in real time, ensuring efficient weed management without manual intervention. Additionally, by integrating edge computing, the ESP32-CAM can process images locally, reducing the need for external servers and improving response time. This makes it an effective solution for smart agriculture, automated plant care, and AI-driven precision farming, helping to enhance crop health and yield.



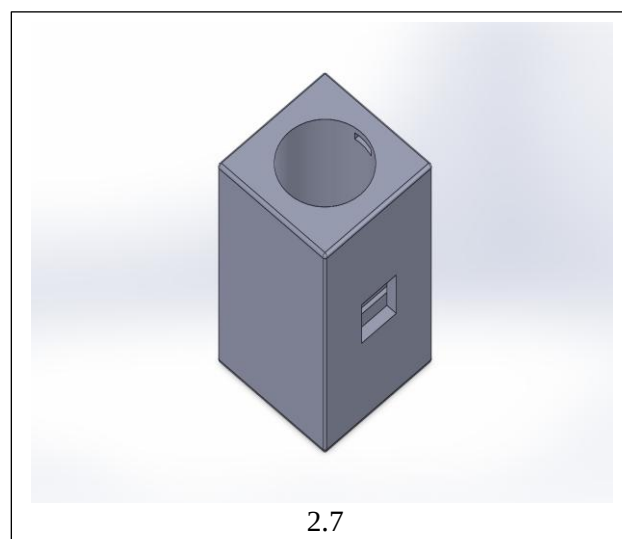
2.6

3.2 3D - Printed Base:

A 3D-printed base for the plant serves as a customized and functional support structure that enhances the stability, aesthetics, and smart features of the plant monitoring system. The base can be designed using CAD software like **SolidWorks**, Fusion 360, or CATIA and then fabricated using FDM (Fused Deposition Modeling) or SLA (Stereolithography) 3D printing technologies.

The base design can include built-in compartments to house electronic components such as the ESP32 module, sensors (moisture, temperature, humidity), and battery/power supply. Additionally, it can feature an integrated facial display to give expressive feedback on the plant's health, making the system interactive. The base can also support wire management channels to keep the circuit neat and safe from water exposure. Using lightweight and durable materials like PLA or PETG, the 3D-printed base ensures strength and weather resistance, making it suitable for indoor and outdoor use.

Furthermore, the customizability of 3D printing allows for the integration of self-watering systems, modular sensor attachments, and artistic designs to enhance both functionality and visual appeal. This approach not only improves the plant's care but also supports smart agriculture innovations and sustainable design practices.

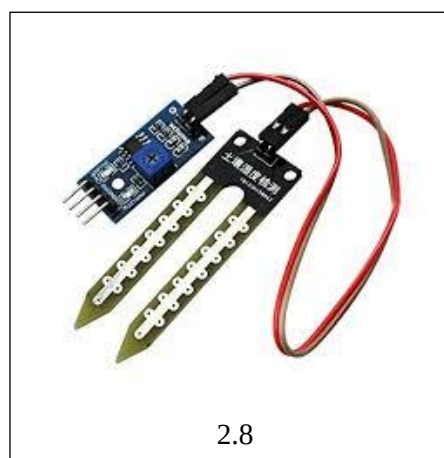


3.2.1 REASONS FOR DESIGN:

The 3D-printed base is designed to provide a stable, functional, and aesthetically appealing support system for the smart plant monitoring system. One of the main reasons for this design is its ability to integrate electronic components like the ESP32 module, sensors, and battery compartments in a compact and organized manner, ensuring protection from environmental factors like moisture and dust. The base also allows for customized mounting options for the ESP32-CAM, ensuring an optimal camera angle for AI-based weed detection and plant health monitoring.

Additionally, 3D printing offers high flexibility in design, allowing for the incorporation of self-watering systems, modular sensor attachments, and wire management channels for a clean and efficient setup. The use of lightweight yet durable materials such as PLA, PETG, or ABS ensures weather resistance, sustainability, and long-term durability for both indoor and outdoor applications. Furthermore, the base includes an interactive facial display, enhancing user engagement by providing visual feedback on plant health. This innovative design improves functionality, durability, and user experience, making it an essential component of the smart plant monitoring system.

3.3 SOIL MOISTURE SENSOR:



A Soil Moisture Sensor is a crucial component in plant monitoring and smart agriculture systems, used to measure the water content in the soil and ensure optimal

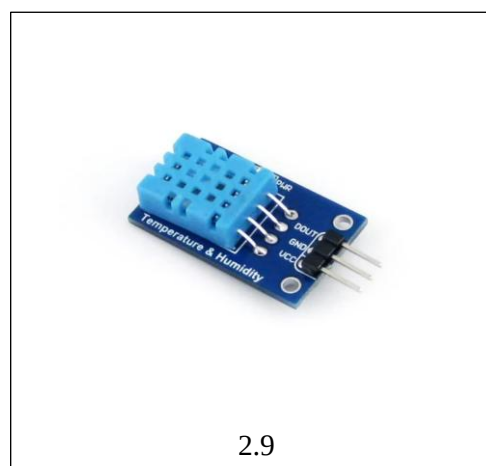
plant growth. It works by detecting the electrical conductivity or capacitance of the soil, which changes based on its moisture level. The sensor consists of two probes that are inserted into the soil, and as moisture increases, the conductivity improves, giving a higher reading. Conversely, dry soil results in lower conductivity and a lower sensor output.

There are two common types of soil moisture sensors:

1. **Resistive (Analog) Sensors** – These measure the resistance between two probes. More moisture reduces resistance, indicating higher soil moisture.
2. **Capacitive Sensors** – These are more durable as they **prevent corrosion** by measuring soil moisture through **capacitive sensing**, which detects changes in the dielectric constant of the soil.

The ESP32 module can read the sensor's output and process the data to determine whether the plant needs watering. If the moisture level drops below a predefined threshold, the system can trigger alerts on a mobile app, display expressions on a facial display, or even activate an automatic irrigation system. This automation enhances plant health management, prevents overwatering or underwatering, and optimizes water conservation, making it ideal for smart plant care systems.

3.4 TEMPERATURE SENSOR:



The DHT11 is a low-cost digital sensor used to measure temperature and humidity, making it ideal for plant monitoring and smart agriculture applications. It features a thermistor to detect temperature and a capacitive humidity sensor to measure moisture levels in the air. The sensor provides digital output, which can be easily read using the ESP32 module for real-time monitoring and automation.

Key Features of DHT11:

- **Temperature Range:** 0°C to 50°C ($\pm 2^\circ\text{C}$ accuracy)
- **Humidity Range:** 20% to 90% RH ($\pm 5\%$ accuracy)
- **Operating Voltage:** 3.3V - 5V (compatible with ESP32)
- **Digital Output:** Uses **single-wire communication** for easy integration
- **Low Power Consumption:** Ideal for battery-operated projects

Working Principle:

The DHT11 sensor continuously monitors the surrounding environment. The humidity sensor measures moisture using capacitance changes, while the thermistor detects temperature variations. This data is sent to the ESP32, which can process the readings and trigger actions like alerts, irrigation, or climate control adjustments.

Applications in Plant Monitoring:

- Helps regulate **temperature-sensitive plants** by triggering fans or heaters
- Maintains optimal humidity levels for **healthy plant growth**
- Sends real-time data to a **mobile app** for remote monitoring
- Improves **water conservation** by adjusting irrigation based on environmental conditions

3.5 LDR SENSOR:



An LDR (Light Dependent Resistor) sensor is a key component in plant monitoring systems, used to measure light intensity and ensure plants receive optimal sunlight for growth. The sensor works by changing its resistance based on light exposure—when light levels are high, the resistance decreases, and when it's dark, the resistance increases. This allows the ESP32 module to interpret light levels and make decisions accordingly.

Key Features of LDR Sensor:

- Detects light intensity for monitoring plant sunlight exposure
- Analog Output that can be read by an ESP32's ADC (Analog-to-Digital Converter)
- Low Power Consumption, making it ideal for continuous monitoring
- Simple Integration with ESP32 and other microcontrollers

Working Principle:

The LDR sensor is made of a semiconductor material whose resistance changes based on the amount of light falling on it. The ESP32 reads the sensor's resistance value and converts it into a light intensity level. This data can be used to trigger actions, such as:

- Sending alerts to a mobile app if a plant is not receiving enough sunlight

- Activating an artificial grow light if sunlight is insufficient
- Adjusting smart shading systems to prevent excessive sunlight exposure

Applications in Smart Plant Monitoring:

- Ensures plants get the right amount of sunlight
- Prevents overexposure to harsh sunlight
- Automates light-based adjustments for indoor plants
- Helps in greenhouse automation for climate control

3.6 I2C DISPLAY:



An I2C display is a low-power, compact, and easy-to-use display module that communicates with microcontrollers like the ESP32 using the I2C (Inter-Integrated Circuit) protocol. In a smart plant monitoring system, the I2C display can be used to show real-time data such as soil moisture, temperature, humidity, light levels, and AI-based weed detection results. It can also display a facial expression to indicate the plant's health status, making the system interactive.

Key Features of I2C Display:

- Uses I2C communication, requiring only two pins (SDA and SCL), reducing wiring complexity
- Common types: OLED (Organic Light Emitting Diode) and LCD (Liquid Crystal Display)
- Low Power Consumption, making it suitable for battery-powered devices
- High Contrast & Visibility, even in bright environments
- Compact Size, ideal for integrating into a 3D-printed plant base

Working Principle:

The I2C display module connects to the ESP32 via the SDA (data) and SCL (clock) lines. The ESP32 sends data to the display using the I2C protocol, allowing it to show real-time sensor readings and alerts. The display can be programmed to:

- Show plant health data such as moisture level, temperature, and humidity
- Display expressive faces (happy, sad, neutral) based on plant condition
- Indicate low water levels or the need for light exposure adjustments

Applications in Smart Plant Monitoring:

- Provides real-time data without needing a mobile app
- Enhances interactivity with a facial display feature
- Helps in visualizing sensor readings at a glance
- Reduces wiring complexity compared to traditional LCD displays.

3.6.1 REASON FOR DISPLAY:

The display is integrated into the plant monitoring system to provide real-time information on environmental conditions without requiring a mobile app. It allows users to quickly check important data such as soil moisture, temperature, humidity, and light levels at a glance. This is particularly useful in indoor gardens, smart agriculture, and automated plant care systems, where instant feedback is crucial for maintaining optimal plant health. Additionally, a display reduces dependency on

wireless connectivity, ensuring the system remains functional even if the mobile app is unavailable.

3.6.2 USAGE OF DISPLAY:

The display is used to show sensor readings, alerts, and AI-based weed detection results. It can also feature facial expressions (happy, sad, neutral) to visually indicate the plant's health, making the system more interactive and user-friendly. Mounted on a 3D-printed base, the display provides easy access to plant status updates, helping users make timely adjustments. In automated irrigation systems, it can show alerts when the plant needs watering or when light exposure is insufficient. The display's ability to present real-time AI analysis and sensor data makes it an essential component for efficient and smart plant care.

3.7 LITHIUM - ION BATTERY:



A Lithium-Ion (Li-Ion) battery is a rechargeable power source widely used in portable electronics, IoT devices, and smart systems, including plant monitoring systems. It is preferred for its high energy density, lightweight design, and long lifespan compared to traditional batteries. Li-Ion batteries operate by moving lithium ions between the anode (graphite) and cathode (lithium metal oxide) through an electrolyte during charging and discharging cycles.

Key Features of Li-Ion Battery:

- High Energy Density: Stores more power in a compact size
- Rechargeable & Long Lifespan: Supports hundreds to thousands of charge cycles
- Low Self-Discharge Rate: Retains power for longer periods
- Lightweight & Compact: Ideal for portable and embedded systems
- Provides Stable Voltage Output: Ensures reliable performance for microcontrollers

Usage in Smart Plant Monitoring Systems:

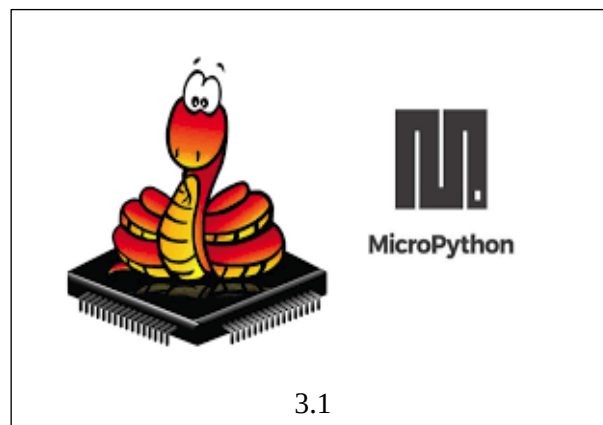
In smart plant monitoring, Li-Ion batteries are used to power ESP32 modules, sensors, and displays. Their low power consumption and rechargeable capability make them ideal for long-term remote monitoring. They enable the system to function off-grid, allowing for wireless deployment in outdoor and greenhouse environments. Additionally, Li-Ion batteries can be combined with solar charging circuits for sustainable and eco-friendly power solutions in precision agriculture and IoT-based plant care.

SOFTWARE COMPONENTS

3.8 MICROPYTHON FOR ESP - 32

3.8.1 What is a Micro python?

Micro Python is a lightweight implementation of Python 3 designed to run on microcontrollers and embedded systems with limited resources. It provides a Python-like programming environment while being optimized for low-power and low-memory devices, making it ideal for IoT applications, robotics, and automation projects.



3.8.2 Installing Micro python and Setting Up ESP32

Step 1: Download and Install Thonny (or uPyCraft IDE)

1. Download Thonny IDE from the official site: <https://thonny.org/>.
2. Alternatively, you can use uPyCraft IDE for Micro Python development.
3. Install the software and launch Thonny IDE.

Step 2: Download and Flash Micro Python Firmware

1. Visit the official Micro Python ESP32 firmware page: <https://micropython.org/download/esp32/>.
2. Download the latest .bin file for ESP32.

3. Install esptool (required for flashing firmware):

Open a terminal or command prompt and run:

```
pip install esptool
```

4. Connect the ESP32 board to your computer via USB and check the COM port.

5. Erase the ESP32 flash memory using the following command:

```
esptool.py --chip esp32 erase_flash
```

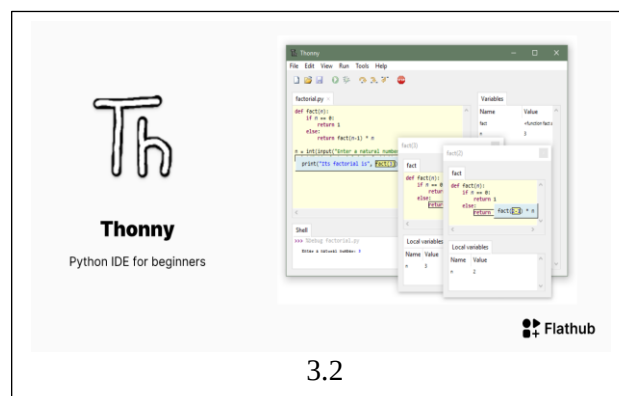
6. Flash the Micro Python firmware to ESP32 using:

```
esptool.py --chip esp32 --port COMx write_flash -z  
0x1000 firmware.bin
```

7. (Replace *COMx* with the correct port and *firmware.bin* with the downloaded firmware file name.)

Step 3: Connect ESP32 to Thonny IDE

1. Open Thonny IDE and go to Tools → Options → Interpreter.
2. In the Interpreter dropdown, select Micro Python (ESP32).
3. In the Port section, select the correct ESP32 COM port.
4. Click OK and restart Thonny.



Step 4: Test Micro Python on ESP32

1. Open the **Shell** in Thonny.
2. Type the following Micro Python command to test LED blinking:

```
from machine import Pin

import time

led = Pin(2, Pin.OUT) # GPIO2 is the built-in LED

while True:

    led.value(not led.value())

    time.sleep(1)
```

3. Click **Run** to execute the code. If the **LED blinks**, Micro Python is successfully installed!

3.8.3 Programming ESP32 in Micro python:

Micro Python makes ESP32 programming simple and efficient, allowing users to write Python-based scripts for embedded systems and IoT applications. It provides a lightweight runtime environment that supports GPIO control, WIFI, I2C, SPI, PWM, and more.

1. Setting Up Micro Python on ESP32:

Before programming, ensure that Micro Python is installed on the ESP32. If not, follow these steps:

1. **Flash Micro Python firmware** onto ESP32 using [esptool.py](#).
2. Use **Thonny IDE** or **uPyCraft IDE** to communicate with the ESP32.

3.9 APPLICATIONS FOR INTERNET OF THINGS (IOT):

The Internet of Things (IoT) enhances the efficiency and automation of the smart plant monitoring system by enabling real-time data collection, remote monitoring, and automated actions. ESP32-based sensors continuously track soil moisture, temperature, humidity, and light levels, transmitting data to a mobile app or cloud platform for real-time insights. The AI-powered ESP32-CAM module captures images to detect weeds and plant diseases, alerting users to take necessary actions. IoT integration enables an automated irrigation system, where a relay-controlled water pump activates when soil moisture drops below a threshold, reducing manual effort and water wastage. A facial expression display on the plant pot visually represents the plant's health, making it interactive and user-friendly. The system logs historical data for trend analysis, sending notifications and alerts when unfavorable conditions arise. Additionally, energy-efficient components like Li-Ion batteries and solar charging ensure sustainability. This IoT-enabled approach makes home gardening, greenhouse management, and precision farming smarter, more efficient, and fully automated.

3.10 TENSORFLOW



TensorFlow is an open-source machine learning (ML) and deep learning framework developed by Google. It provides a flexible and scalable platform for building and deploying AI models, making it widely used in computer vision, natural language processing (NLP), robotics, and IoT applications. TensorFlow supports multiple

programming languages, including Python, C++, and JavaScript, and is optimized for both CPU and GPU acceleration.

Key Features of TensorFlow:

- Deep Learning & Neural Networks: Supports CNNs, RNNs, and transformers.
- TensorFlow Lite (TF-Lite): Optimized for edge devices like ESP32, Raspberry Pi, and mobile apps.
- Pre-trained Models: Offers models for image classification, object detection, and speech recognition.
- Scalability: Can run on cloud servers, desktop PCs, and microcontrollers.
- Keras Integration: Uses Keras API for easy model development.

TensorFlow in Smart Plant Monitoring & Weed Detection

TensorFlow can be used to develop AI-based weed detection in this project. The ESP32-CAM captures images of plants, and a TensorFlow-trained model classifies them as healthy plants or weeds. TensorFlow Lite allows deployment on low-power edge devices, ensuring real-time detection. The AI model can also detect plant diseases from leaf images, helping in precision agriculture and automated farming solutions.

3.11 GOOGLE COLAB:



Google Colab (Collaboratory) is a free cloud-based Jupyter notebook environment that allows users to write and execute Python code for machine learning, deep learning, and data science. It is hosted by Google and provides free access to GPUs and TPUs, making it an excellent platform for training AI models without requiring powerful local hardware.

Key Features of Google Colab:

- **Free GPU & TPU Acceleration:** Supports NVIDIA GPUs and Google TPUs for fast computation.
- **Pre-installed Libraries:** Comes with TensorFlow, Keras, OpenCV, PyTorch, and NumPy pre-installed.
- **Cloud Storage Integration:** Easily connects to Google Drive for saving and accessing files.
- **Supports Collaboration:** Multiple users can edit and run the same notebook in real time.
- **No Installation Required:** Runs entirely in a web browser, eliminating setup issues.

Google Colab can be used to train an AI model using TensorFlow for weed and plant disease detection. The workflow involves:

1. Uploading a dataset of plant images (healthy plants and weeds).
2. Training a CNN model using TensorFlow/Keras.
3. Exporting the trained model as a TensorFlow Lite (TF-Lite) file for deployment on ESP32-CAM.

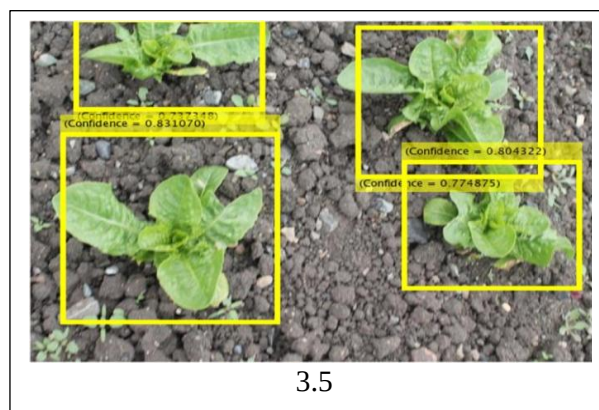
3.12 APPLICATION OF WEED DETECTION AND PLANT DISEASE AI:

Weed detection using artificial intelligence (AI) helps in precision farming by identifying and differentiating unwanted plants from crops. Using an ESP32-CAM module, real-time images of plants are captured and processed through a pre-trained

machine learning model using TensorFlow Lite. The model classifies whether a plant is a crop or a weed, enabling targeted weed removal. This reduces the need for manual labor and excessive herbicide use, making farming more efficient and eco-friendly. The AI system provides real-time alerts via a mobile app, allowing users to take immediate action. By integrating AI with IoT, the system ensures automated weed detection, improving crop yield and farm productivity.

AI-based plant disease detection helps in early identification and prevention of infections that affect crop health. The system captures images using the ESP32-CAM and analyzes them with a TensorFlow-trained model to detect diseases based on leaf color, texture, and patterns. The AI can classify various plant diseases and suggest possible treatments. This reduces crop loss, minimizes chemical pesticide usage, and helps in sustainable farming. Users receive real-time notifications about plant health, allowing timely intervention. By combining AI, IoT, and machine learning, the system ensures a data-driven approach to plant care, benefiting home gardeners, greenhouse monitoring, and large-scale agriculture.

3.13 About Weed Detection (AI)



Weed detection using Artificial Intelligence (AI) is a smart farming solution that helps in identifying and classifying weeds among crops. By utilizing machine learning models and image processing techniques, AI can differentiate between weeds and healthy plants with high accuracy. This system reduces the reliance on manual weeding and excessive herbicide use, leading to more efficient and eco-

friendly farming practices. The ESP32-CAM module, combined with TensorFlow Lite, captures real-time images, processes them, and provides automated weed detection for precision agriculture.

3.14 About Plant Disease (AI)



3.6

AI-powered plant disease detection is an advanced method to identify infections, nutrient deficiencies, and abnormalities in plants using image processing and deep learning. By analyzing leaf color, shape, and texture, AI can predict plant diseases at an early stage, allowing farmers to take preventive action. The ESP32-CAM captures plant images, and a trained TensorFlow model classifies diseases, sending real-time alerts to the user. This system helps in reducing crop loss and improving agricultural efficiency.

3.15 Internet of Things (IoT) for Plant Monitoring

The Internet of Things (IoT) plays a crucial role in modern plant monitoring systems, enabling real-time tracking of plant health parameters like soil moisture, temperature, humidity, and light intensity. By integrating sensors with IoT-enabled devices such as ESP32, data is collected, processed, and transmitted to a mobile app or cloud platform. This system allows remote monitoring, automated control, and intelligent decision-making for better plant growth. Additionally, AI-powered weed detection and plant disease identification enhance the efficiency of plant care, making the system highly advanced and efficient.

CHAPTER - 4

Results and Discussions

The IoT-based Plant Monitoring System integrates sensors, AI-powered weed detection, plant disease identification, and a facial display interface to provide real-time plant health updates. The system consists of an ESP32 microcontroller, ESP32-CAM, sensors (DHT11, soil moisture, LDR, etc.), and a 3D-printed plant base with a facial display.

1. Data Collection:

- Soil moisture sensor measures water levels in the soil.
- DHT11 sensor records temperature and humidity.
- LDR sensor detects light intensity for plant growth analysis.
- ESP32-CAM captures images of plants for weed and disease detection.

2. Data Processing & Transmission:

- The ESP32 microcontroller gathers sensor data and transmits it via Wi-Fi to a cloud platform.
- AI models (TensorFlow Lite) analyze plant images for weed and disease detection.
- The processed data is displayed on a mobile app for real-time monitoring.

3. Automated Actions & Alerts:

- If the soil is dry, an alert is sent to the user for watering.
- If plant diseases or weeds are detected, recommendations are given via the app.
- The 3D-printed facial display in the pot changes expressions based on plant health status.

4. User Interaction & Control:

- Users can monitor plant health remotely through an IoT dashboard.
- The system sends notifications for watering, light conditions, or disease detection.
- Future enhancements include automated irrigation and robotic weed removal.

The project successfully demonstrated an IoT-based plant monitoring system with AI-powered weed detection and plant disease identification. The system effectively collects, processes, and transmits plant health data to a mobile app, allowing remote monitoring and real-time decision-making.

Key Observations:

- **Sensor Accuracy:**
 - Soil moisture readings provided precise water level information.
 - The DHT11 sensor effectively tracked temperature and humidity.
 - The LDR sensor successfully detected variations in sunlight.
- **AI-based Weed & Disease Detection:**
 - The ESP32-CAM with TensorFlow Lite correctly identified weed presence and plant diseases in test samples.
 - Accuracy improved with better dataset training, reducing false positives.
- **IoT Data Transmission & Alerts:**
 - The ESP32 sent real-time data to the cloud and mobile app.
 - The system provided timely alerts for watering, light conditions, and disease detection.
- **3D-Printed Facial Display:**
 - The interactive facial display changed expressions based on plant health, enhancing user engagement.

Challenges & Future Improvements:

- Enhancing AI model accuracy with a larger dataset for weed and disease detection.
- Implementing automated irrigation and weed-removal mechanisms for full automation.
- Improving battery efficiency with solar-powered charging.

CONCLUSIONS

The IoT-based plant monitoring system successfully integrates sensors, AI-powered weed detection, plant disease identification, and an interactive facial display to improve plant care. By using ESP32 and ESP32-CAM, the system provides real-time data on soil moisture, temperature, humidity, and light intensity, ensuring optimal plant growth.

Key Achievements:

- ✓ **Automated plant monitoring with IoT and AI.**
- ✓ **Accurate weed detection using TensorFlow Lite on ESP32-CAM.**
- ✓ **Real-time alerts & notifications via a mobile app.**
- ✓ **Interactive 3D-printed facial display for plant health status.**
- ✓ **Scalable and cost-effective solution for smart gardening.**

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PO & PSO Attainment

PO	Graduate Attribute	Attained	Justification
PO 1	Engineering knowledge	Yes	Implemented 3D Printing and ESP - 32
PO 2	Problem analysis	Yes	Communication through wireless medium
PO 3	Design/Development of solutions	Yes	Framework of a Plant Base
PO 4	Conduct investigations of complex problems	Yes	Analysis of weed and plant disease detection
PO 5	Modern Tool usage	Yes	Used ESP CAM - 32
PO 6	The Engineer and society	No	
PO 7	Environment and Sustainability	Yes	Improves Pollution control and Gardening
PO 8	Ethics	No	
PO 9	Individual and team work	Yes	Comprehensive team effort for fabrication

PO	Graduate Attribute	Attained	Justification
PO 10	Communication	No	
PO 11	Project management and finance	Yes	Budget friendly design
PO 12	Life-long learning	No	
PSO 1	To analyse, design and develop solutions by applying the concepts of Robotics for societal and industrial needs.	Yes	The integration of the plant monitoring , weed detection and plant disease make the healthy growth of plants.
PSO 2	To create innovative ideas and solutions for real time problems in Manufacturing sector by adapting the automation tools and technologies.	Yes	The Plant monitoring makes the plant to be maintain water wastage and proper growing so it's easier to manage.